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NARULA INSTITUTE OF TECHNOLOGY
AN AUTONOMOUS INSTITUTE UNDER MAKAUT
2025
2024-25 EVEN SEMESTER
CH(IT)201 - ENGINEERING CHEMISTRY

TIME ALLOTTED: 3Hours**Maximum Marks : 70****Instructions to the candidate:***Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. If ideal gas undergoes isothermal compression then internal energy change & enthalpy change must be (1) C04 BL1 PO1
 - a. Negative
 - b. Positive
 - c. None
 - d. Zero
2. Reaction of Propene with HBr in presence of Peroxide follows (1) C01 BL2 PO2
 - a. Nucleophilic addition path
 - b. Electrophilic addition path
 - c. Electrophilic substitution path
 - d. Free radical mechanism
3. Fuel cells convert: (1) C04 BL1 PO1
 - a. Mechanical energy to electrical
 - b. Electrical to chemical
 - c. Chemical to electrical
 - d. Thermal to electrical

4. The anodic area in corrosion acts as: (1) C04 BL1 PO1
 a. Reduction site
 b. Oxidation site
 c. Neutral site
 d. Protective site
5. The polymer with the highest thermal stability is: (1) C02 BL1 PO1
 a. Teflon
 b. Nylon
 c. Polyethylene
 d. Polystyrene
6. Mg and Li share similar properties due to (1) C01 BL1 PO1
 a. Belong to same group
 b. Diagonal relationship
 c. Same electron affinity
 d. Both have same IP
7. Which one is the correct statement? (1) C01 BL1 PO1
 (a) O_2 is diamagnetic but N_2 is paramagnetic (b) O_2 is paramagnetic but N_2 is diamagnetic
 (c) both of these are true (d) none of these are true
8. Correct increasing order of acidity is- (1) C01 BL1 PO1
 (a) $CH_4 < NH_3 < H_2O < HF$ (b) $NH_3 < H_2O < HF < CH_4$
 (c) $H_2O < HF < NH_3 < CH_4$ (d) $HF < NH_3 < CH_4 < H_2O$
9. Inversion of configuration is associated with which of the following? (1) C04 BL1 PO1
 (a) $E1$ reaction (b) S_N2 reaction
 (c) S_N1 reaction (d) free radical halogenations
10. ΔH for an exothermic reaction is (1) C04 BL2 PO1
 (a) positive (b) negative
 (c) zero (d) none of these
11. Which of the following equation is incorrect? (1) C04 BL3 PO1
 (a) $H = G + TS$ (b) $G = H + TS$
 (c) $A = U - TS$ (d) $H = U + PV$
12. Germanium is an example of (1) C01 BL1 PO1

- (a) Intrinsic semiconductor (b) P-type semiconductor
(c) N-type semiconductor (d) An insulator

Part B**(Answer any three of the following) 3x5=15**

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|--|-----|-----|-----|-----|
| 13. Why the first ionization potential of the elements ns^2np^1 and ns^2np^4 configurations do not follow any regular trend? | (5) | C01 | BL1 | PO1 |
| 14. Write differences between n-type and p-type semiconductors with example. | (5) | C03 | BL1 | PO1 |
| 15. What is tacticity? Classify the polymers with respect to tacticity with diagram. | (5) | C02 | BL1 | PO1 |
| 16. Efficiency of a heat engine is always less than unity; explain in the light of Carnot's Cycle. | (5) | C02 | BL2 | PO1 |
| 17. On which factors SN_2 reaction rate depend? Discuss in details. | (5) | C01 | BL1 | PO2 |

Part C**(Answer any three of the following) 3x15=45**

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| 18. Answer the followings. | | | | |
| (a) Show the mechanism of SN_1 reaction with proper example. What are the factors on which SN_1 reaction rate depend also discuss every factor in details. | (5) | C01 | BL1 | PO2 |
| (b) What are the different types of electronic transitions in UV spectroscopy? Discuss in details. | (5) | C05 | BL2 | PO2 |
| (c) Propene + HBr in presence of Peroxide = Products? Discuss mechanism. | (5) | C01 | BL1 | PO2 |
| Propene + HBr = Products? Discuss mechanism. | | | | |
| 19. a. Explain the Bohr's theory for a one-electron system and derive an expression for the energy of an electron in an orbit. | (15) | C01 | BL2 | PO1 |
| 3+5 | | | | |
| b. What is ionization potential? On which factors ionization potential depend? 2+5 | | | | |
| 20. (a) What are the applications of nano materials in Chemistry? | (5) | C02 | BL1 | PO1 |
| (b) Write differences between thermoplastic and thermosetting polymers with example. | (5) | C04 | BL1 | PO1 |

- (c) What is Absorbance? Provide schematic graphical presentation of UV-Vis and IR spectra (5) C05 BL1 PO1
21. a) Explain enthalpy change for exothermic reactions and endothermic reactions using graph. (15) C04 BL1 PO2
- b) Define entropy with significance. Mention criteria of spontaneity for a thermodynamic process.
- c) Calculate the emf of the following cell at 298 K. $\text{Cu (s)} | \text{Cu}^{2+} (0.130 \text{ M}) || \text{Ag}^+ (1.00 \text{ M}) | \text{Ag (s)}$. Given reduction potential: $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$ and $E^\circ_{\text{Ag}^+/\text{Ag}} = +0.80 \text{ V}$.
22. (a) Write the physical concept of Entropy. How entropy changes for the following reactions: (5) C04 BL3 PO1
- (a) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (b) $\text{NaCl}(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
- (b) Predict the feasibility of the given reaction with proper explanation (5) C04 BL4 PO1
- $2\text{Ag}^+ + \text{Cu}^{2+} \rightarrow 2\text{Ag} + \text{Cu}$
- Given:
- $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$
- $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$
- (c) What are the consequences of Schotky defect? What kind of defects are shown by AgBr. (5) C04 BL3 PO1